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Protocol generation date:
No.:

LABORATORY TEST PROTOCOL

Patient code:
Age:
Sex:

Sample type: dried blood spot (DBS)
Sample collected:
Sample received:

Test	Analysis date	Result	Measurement units	Interpretation	Method	Employee license No.
Vitamin D			nmol/l		3	

Analysis method:

3 – Liquid chromatography–mass spectrometry (LC-MS/MS)

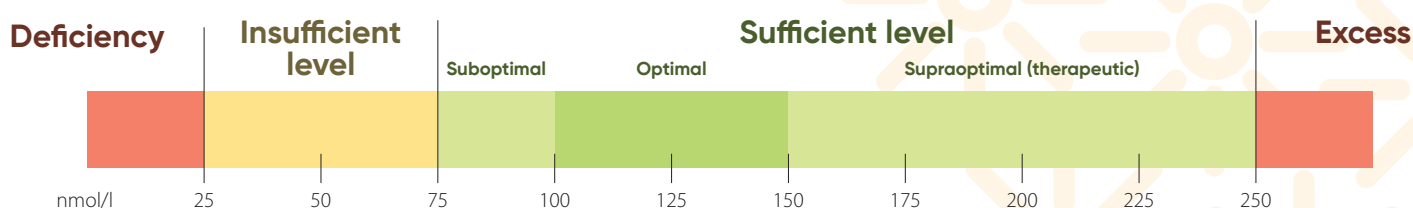
Hey there, it's awesome to have you here!

Understanding your vitamin D level is a fantastic step toward feeling your best! This guide is your key to understanding what those results mean and how you can boost your vitamin D status for optimal health.

Fun fact alert! Vitamin D is way cooler than just a vitamin. It's actually a fat-soluble hormone, found in two main forms: cholecalciferol (D3 – animal-sourced) and ergocalciferol (D2 – comes from plant sources and fortified foods) [1,2].

Wondering where your vitamin D level stands? Let's shine a light on your results.

Here are the reference ranges for vitamin D levels:



Why keeping sufficient vitamin D levels is key

Don't underestimate the power of vitamin D! Maintaining sufficient levels of vitamin D is crucial for several reasons. Firstly, it facilitates the absorption of phosphorus and calcium in the intestines, ensuring that our bones remain strong and healthy [3]. Without enough vitamin D, children can develop rickets, leading to weakened bones, while adults may experience osteomalacia, resulting in low bone density and muscle weakness [4]. Moreover, long-term deficiency can contribute to conditions like osteoporosis and increase the risk of various diseases such as colon cancer, cardiovascular diseases, and diabetes mellitus [5]. Moreover, clinical studies link vitamin D deficiency to an increased risk of blood clots, suggesting a potential role for vitamin D in regulating thrombosis-related pathways [6]. Shockingly, studies reveal that approximately 1 billion people worldwide are affected by vitamin D deficiency, with prevalence rates ranging from 20% to 90% in regions like Europe, the USA, and the Middle East [5].

Vitamin D: The hidden edge for peak performance

Research suggests vitamin D plays a crucial role in muscle function, potentially improving strength, power, and recovery [7,8]. Studies even suggest it might help manage weight by regulating leptin, the satiety hormone, potentially reducing cravings and aiding weight management goals [9].

Fascinating insights in the world of scientific discovery

Beyond its well-characterized physiological functions, vitamin D has emerged as a potential player in numerous other health and disease processes. Notably, research suggests it may delay cellular senescence through the reduction of oxidative stress [10]. In biology, cellular senescence is the process by which a cell undergoes irreversible replicative arrest (cell ages and permanently stops dividing but does not die), a well-established hallmark of biological aging and associated diseases [11,12].

Intriguingly, these findings position vitamin D as a potential key to unlocking longevity and combating age-related diseases.

Beyond physical benefits, the link between vitamin D and mental well-being is gaining significant attention. Vitamin D receptors are present throughout the brain, including areas critical for mood regulation, memory, and learning (specifically in the prefrontal cortex, hippocampus, cingulate gyrus, thalamus, hypothalamus, and substantia nigra) [13]. Studies have shown that the active form of vitamin D's ability to penetrate the blood-brain barrier underscores its role in controlling neurotransmitters such as serotonin and melatonin, further highlighting its relevance in mental well-being, particularly in regulating mood and sleep patterns [13,14]. Consequently, individuals with vitamin D deficiency may experience symptoms such as mental fog, depression, anxiety, and mood swings [13–15]. Interestingly, a recent study suggests that people with lower vitamin D levels have an increased risk of developing dementia earlier in life, compared to those with normal levels [16].

Silent Threat: Vitamin D Deficiency

Vitamin D deficiency often goes unnoticed as it may not cause immediate symptoms, but can manifest over time. Moreover, certain factors can increase the risk of deficiency, including:

- Skin type: Darker skin requires more sunlight for vitamin D synthesis.
- Insufficient sun exposure: Living in northern latitudes, indoor work, or night shifts may reduce sun exposure [17].
- Age: As we age, our skin becomes less efficient at making vitamin D, and lifestyle changes can lead to less sun exposure.
- Weight: Higher body fat can limit vitamin D absorption [18].
- Gut Health: Conditions affecting fat absorption can also impact vitamin D levels.
- Specific patient populations, such as those with chronic renal failure undergoing hemodialysis, renal transplant recipients, individuals with liver disease, or post-liver transplantation patients, exhibit a high prevalence of vitamin D deficiency [19].

Vitamin D deficiency is very difficult to notice, but you should pay attention to the following symptoms:

- Persistent fatigue and sleepiness.
- Mood swings.
- Frequent infections and colds.
- Slow wound healing.
- Muscle spasms.
- Occasional insomnia and night sweats.
- Increased sensitivity to pain.
- Etc [20].



While vitamin D deficiency can harm your health, the other side of the coin isn't much better. Too much vitamin D, often caused by excessive supplement intake, can lead to vitamin D toxicity. This occurs when excess vitamin D raises blood calcium levels (hypercalcemia), causing symptoms like fatigue, nausea, and confusion. In severe cases, kidney problems, heart rhythm issues (arrhythmia), and difficulty walking (ataxia) may develop [20,21]. Keep in mind: moderation is the secret ingredient for a balanced approach, especially when it comes to your vitamin D intake!

Let's shed some light on how to naturally increase your vitamin D levels. While information may be limited, there are several effective strategies supported by science. Discover actionable steps to optimize your vitamin D levels and support your overall health.

Considering that an impressive 90% of our body's vitamin D comes from sunlight, with the remaining 10% obtained through diet, it's no surprise that the primary strategies for increasing vitamin D levels are sunlight exposure and dietary choices.



Our skin acts like a vitamin D factory, using sunlight to convert a compound called 7-dehydrocholesterol into the active form (D3).

Here's the trick: Aim for 5-30 minutes of unprotected sun exposure most days, but prioritize sun protection afterwards to minimize skin cancer risk. Think of it as a short "sun deposit" followed by sunscreen and protective clothing.

Keep in mind: Factors like season, location, and skin tone can all influence your ideal sun exposure time [22].



While sunlight exposure is a key source of vitamin D, not everyone obtains enough due to limited sun exposure or other factors. However, there's no need to worry, as nature provides an array of dietary options to meet our vitamin D needs. Excellent food sources include various fish (5-25 µg/100 g), mushrooms (21.1-58.7 µg/100 g), Reindeer lichen (87 µg/100 g), and fish liver oils (250 µg/100 g). Additional dietary sources include cheese, beef liver, eggs (1.3-2.9 µg/100 g), dark chocolate (4 µg/100 g), and fortified foods like milk, yogurt, fat spreads, orange juice, and breakfast grains [23].



Studies suggest that regular exercise, particularly endurance activities like running, swimming, or cycling, can be beneficial for those with vitamin D deficiency. Here's why:

Increased Vitamin D Levels: Endurance exercise may help raise vitamin D levels in the bloodstream, especially for people who are deficient.

Potential Mechanism: While the exact mechanism is still under investigation, it's believed that exercise affects the way your body stores and uses vitamin D [24].



While sunlight, diet, and exercise are natural ways to optimize your vitamin D levels, sometimes supplementation might be necessary. However, it's important to remember that the ideal dosage can vary depending on individual factors. Consulting your doctor is crucial to determine the right amount of vitamin D supplementation for you to reach optimal levels and ensure safety.

It was a pleasure sharing this with you. Take care of yourself and keep your vitamin D in check!

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